In this assignment, you will be evaluated on the entire process of transforming abstract client requirements into a concrete system and delivering high-quality code via GitHub.

1. Client Requirements (Vibe to Spec)

Analyze the following "client's thought" and proceed with the implementation.

Client’s Thought:

"Lately, I've been struggling because my work and private tasks are all mixed up. I want to be able to quickly check things that are due soon, and it would be amazing if I could see them in a calendar. Also, I hate tedious registration processes, but I can't afford to lose my data. Oh, and I want to use it instantly when I'm in a hurry, so it has to run really smoothly!"

Functional Requirements from an Engineer's Perspective

Translate the thoughts above into "functional requirements" at a level ready for development.

(Example: "Maintain this data in the DB, and display/control it in this specific way," etc.)

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2. Implementation Requirements

Build a web application equipped with the following features:

User Authentication: Registration and login functionality using email, name, and password.

Task Registration: A card-style UI (Content, Start Date, Due Date, Color Tags).

Dashboard:

View by Tag: A list organized by category.

Calendar View: A visual display on a calendar UI or a list sorted by proximity to the deadline.

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3. Technical Requirements (Performance & Quality)

This project emphasizes User Experience (UX). The following metrics will be part of the evaluation:

Lighthouse Score Measurement:

Performance Score: 90+ is recommended.

Bonus Points: +10 points if **LCP (Largest Contentful Paint)** is **1.2 seconds or less**.

Advanced Task (Bonus):

PWA Support: Additional points will be awarded if the app works offline and is installable.

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4. Quality Assurance (Submission of Test Report)

Even when developing with AI, the final guarantee of operation is the engineer's responsibility. Conduct tests including the following edge cases and report the results.

Mandatory Test Items:

Validation: Does an appropriate error appear when a "past date" is entered as a deadline?

Load & Display: Is the UI still functional and not broken even if 100 tags are registered?

Security: After logging out, is it impossible to view the task list by accessing the URL directly?

Note: Please perform any other tests you deem necessary. Bonus points will be awarded based on the quality and depth of your testing.

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5. Submission Method

Please submit the following two items together in a **Zip file**:

GitHub Repository URL:

The **README.md** must include "Setup Instructions," "Tech Stack Used," and "Key Features/Points of Innovation."

Vercel Link

Test Report:

Include the results of the mandatory test items and a **screenshot of your Lighthouse score**.